3.Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier

# Sample dataset
data = {
    'Outlook': ['Sunny', 'Sunny', 'Overcast', 'Rain', 'Rain', 'Rain',
                'Overcast', 'Sunny', 'Sunny', 'Rain'],
    'Temperature': ['Hot', 'Hot', 'Hot', 'Mild', 'Cool', 'Cool',
                    'Cool', 'Mild', 'Cool', 'Mild'],
    'Humidity': ['High', 'High', 'High', 'High', 'Normal', 'Normal',
                 'Normal', 'High', 'Normal', 'Normal'],
    'Wind': ['Weak', 'Strong', 'Weak', 'Weak', 'Weak', 'Strong',
             'Strong', 'Weak', 'Weak', 'Weak'],
    'Play': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No',
             'Yes', 'No', 'Yes', 'Yes']
}

df = pd.DataFrame(data)

# Encode categorical data
encoder = {}
for col in df.columns:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])
    encoder[col] = le

X = df[['Outlook', 'Temperature', 'Humidity', 'Wind']]
y = df['Play']

# Train ID3 model
model = DecisionTreeClassifier(criterion='entropy')
```

```
model.fit(X, y)

# New sample
sample = pd.DataFrame({
    'Outlook': ['Sunny'],
    'Temperature': ['Hot'],
    'Humidity': ['High'],
    'Wind': ['Weak']
})

# Encode sample
for col in sample.columns:
    sample[col] = encoder[col].transform(sample[col])

# Prediction
prediction = model.predict(sample)

# Convert back to original label
result = encoder['Play'].inverse_transform(prediction)

print("Prediction:", result[0])
```

OUTPUT:

```
Prediction: No
```